

Course Name: Electrical Circuits (2)
Instructors: Dr A. Alzyoud and Eng J. Abu-Ghalune

Date: Jan. 31. 2022
Exam Time: 90 minutes

Prob..	1	2	3	4	5	6	7	8	9	10	11	12	13
Answer													

Prob. 1: Choose the correct answer.

Consider the circuit shown in Figure 1, where the voltage source has $V_s = 5\angle 0^\circ V_{rms}$.

- The open circuit voltage (V_o) between the a-b terminals is
a. $50(1-j1) V$ b. $75(1-j1) V$ c. $25(1-j1) V$ d. None of all
- The short circuit current (I_{sc}) when the a-b terminals are shorted is
a. $-20 A$ b. $-10 A$ c. $-30 A$ d. None of all
- The Thevenin impedance (Z_{th}) between the a-b terminals is
a. $2.5 + j2.5 \Omega$ b. $2.5 - j2.5 \Omega$ c. $-2.5 + j2.5 \Omega$ d. None of all
- The current through a load of $7.5 - j2.5 \Omega$ connected between the a-b terminals is
a. $5(1-j1) A$ b. $10(1-j1) A$ c. $15(1-j1) A$ d. None of all

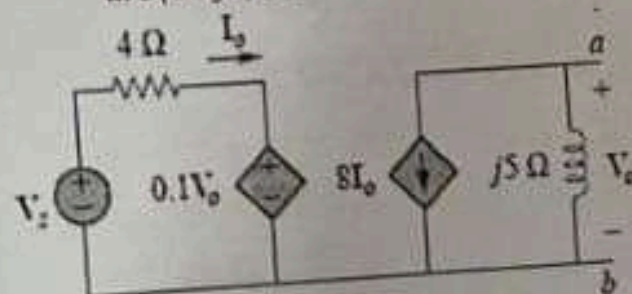


Figure 1

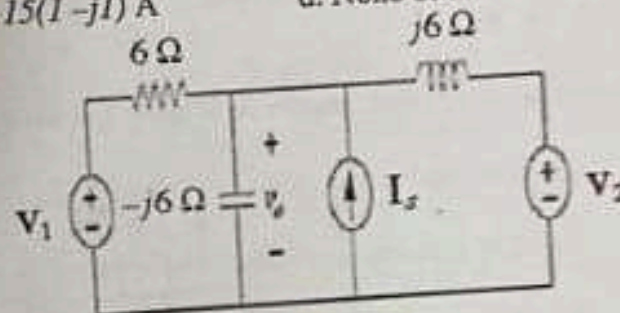


Figure 2

Consider the circuit shown in Figure 2 where $V_1 = 6\angle 0^\circ V_{rms}$, $V_2 = 6\angle 0^\circ V_{rms}$, and $I_s = 1\angle 0^\circ A_{rms}$.

- The voltage across the capacitor due to V_1 source only (in V) is
a. 6 b. 3 c. 12 d. None of all
- The voltage across the capacitor due to V_2 source only (in V) is
a. $-j12$ b. $-j6$ c. $-j3$ d. None of all
- The voltage across the capacitor due to I_s source only (in V) is
a. 6 b. 12 c. 18 d. None of all

Three Loads connected in parallel across a voltage source of $200\angle 0^\circ V_{rms}$, where the loads have:
Load 1: absorbs 250VA at 0.6 lagging p.f., Load 2: absorbs 200W at 0.8 leading p.f., and Load 3: absorbs 300VA and 150VAR.

- The power factor of the load 3 is
a. 0.866 lagging b. 0.707 lagging c. 0.5 lagging d. None of all
- The current of load 1 is
a. $1.3 - j0.75 A$ b. $0.75 - j1.0 A$ c. $1 + j0.75 A$ d. None of all
- The complex power supplied by the source is
a. $564.8 + j501.2 VA$ b. $609.8 + j200 VA$ c. $231.5 + j564.8 VA$ d. None of all
- The power factor of the equivalent load is
a. 0.95 lagging b. 0.748 lagging c. 0.38 lagging d. None of all
- The current supplied by the voltage source in rms is
a. $2.822 - j2.506 A$ b. $2.822 - j1.157 A$ c. $3.049 - j1.0 A$ d. None of all
- The impedance of the equivalent load is
a. $24.87 - j60.67 \Omega$ b. $59.22 + j19.42 \Omega$ c. $60.67 + j24.87 \Omega$ d. None of all

Prob. 1	14	15	16	17	18	19	20	21	22	23	24	25	
Answer													

A balanced Δ -load of $Z_{\Delta} = 8.66 + j5 \Omega$ is supplied by a balanced Y voltages of $V_{as} = 250/\underline{0^\circ}$ V rms in positive sequence.

14. The voltage V_{AB} is
 a. $375 + j216.5$ V
 b. $216.5 + j375$ V
 c. 433 V
 d. None of all
15. The phase current I_{ac} is
 a. $37.5 + j21.65$ A
 b. $j43.3$ A
 c. $-21.65 - j37.5$ A
 d. None of all
16. The line current I_L is
 a. $37.5 + j65$ A
 b. $j75$ A
 c. $37.5 - j65$ A
 d. None of all
17. The total average power supplied by the source is
 a. 48.71 KW
 b. 28.21 KW
 c. 84.37 KW
 d. None of all
18. The total complex power consumed by the load is
 a. $56.48 + j50.12$ KVA
 b. $56.48 + j23.15$ KVA
 c. $48.71 + j28.12$ KVA
 d. None of all

Consider a series RLC circuit has $R = 4 \Omega$, $L = 4$ mH, $C = 0.4 \mu$ F, and $V_s = 12/\underline{0^\circ}$ V rms.

19. The resonant frequency (ω_0) in (rad/sec) is
 a. 25 K
 b. 50 K
 c. 20 K
 d. None of all
20. The quality factor (Q) is
 a. 50
 b. 30
 c. 25
 d. None of all
21. The cutoff frequencies ω_1 and ω_2 (in rad/sec) are
 a. (19.2, 20.8) K
 b. (24, 26) K
 c. (49, 51) K
 d. None of all
22. The voltage across the capacitor (V_C) at resonance (in V) is
 a. 400
 b. 300
 c. 12
 d. None of all

Consider the circuit shown in Figure 3, where $V_s = 15/\underline{0^\circ}$ V rms, $R_1 = 2 \Omega$, and $R_L = 4 \Omega$.

23. The equivalent input impedance (Z_{in}) seen by the source (in Ω)
 a. $4 + j3$
 b. $3 + j3$
 c. $5 + j3$
 d. None of all
24. The current supplied by the source (I_1)
 a. $2 - j1.2$
 b. $1.6 - j1.2$
 c. $2.5 - j2.5$
 d. None of all
25. The power absorbed by the R_L resistor is
 a. 12.5 W
 b. 10.88 W
 c. 8 W
 d. None of all

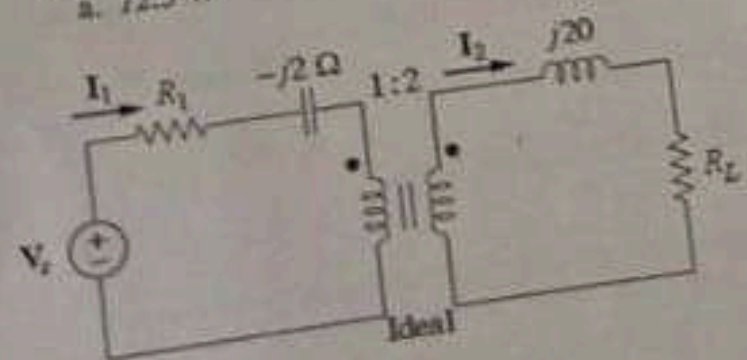


Figure 3

Prob. 1: Choose the correct answer.

1. When the impedance $Z = 8 - j6$ then the voltage across it the current through it by
 a. leads, 36.87° b. lags, 36.87° c. leads, 53.13° d. lags, 53.13°

Consider the circuit shown in Fig. B.1, where $v_s(t) = 150\cos(\omega t + 26.87^\circ)$ and $i(t) = 10\cos(\omega t - 10^\circ)$.

- The equivalent impedance at the terminals of the voltage source is
 a. $16 + j12 \Omega$ b. $8 + j6 \Omega$ c. $20 + j15 \Omega$ d. $12 + j9 \Omega$

3. When $\omega = 1000000000000$ rad/s
 a. 15 mH b. 9 mH c. 9 mH d. 6 mH

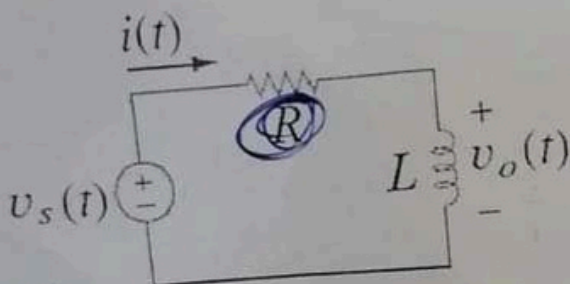


Fig. B.1

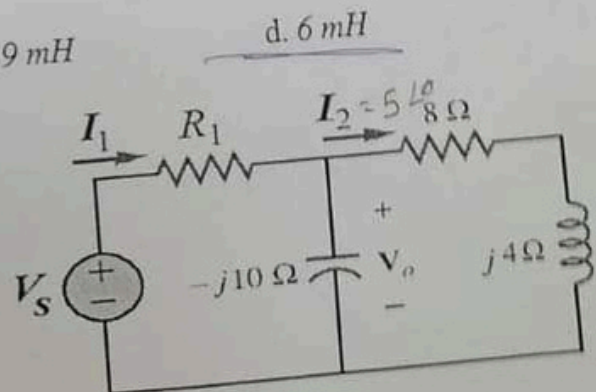


Fig. B.2

Consider the circuit shown in Fig. B.2, where $R_1 = 4 \Omega$.

- The equivalent impedance (Z_{eq}) seen by the source is
 a. $12 - j4 \Omega$ b. $16 - j4 \Omega$ c. $10 - j4 \Omega$ d. $14 - j4 \Omega$
5. When $I_2 = 5/0^\circ \text{ A}$ then $I_1 =$
 a. $6 + j8 \text{ A}$ b. $3 + j4 \text{ A}$ c. $12 + j16 \text{ A}$ d. $9 + j12 \text{ A}$
6. When $V_o = 10/0^\circ \text{ V}$ then $V_s =$
 a. $18 + j4 \text{ V}$ b. $14 + j2 \text{ V}$ c. $16 + j3 \text{ V}$ d. $12 + j1 \text{ V}$

Prob. 1	1	2	3	4	5	6	7	8	9	10	11	12	13
Answer	a		c				c		e	a	b	c	d

Prob. 1: Choose the correct answer.

1. When the impedance $Z = 8 - j6$ then the voltage across it the current through it by
 a. leads, 36.87° b. lags, 36.87° c. leads, 53.13° d. lags, 53.13°

Consider the circuit shown in Fig. B.1, where $v_s(t) = 150\cos(\omega t + 26.87^\circ)$ and $i(t) = 10\cos(\omega t - 10^\circ)$.

- The equivalent impedance at the terminals of the voltage source is
 a. $16 + j12 \Omega$ b. $8 + j6 \Omega$ c. $20 + j15 \Omega$ d. $12 + j9 \Omega$
3. When $\omega = 10000000000 \text{ rad/s}$ = ∞ c. 9 mH d. 6 mH
 a. 15 mH

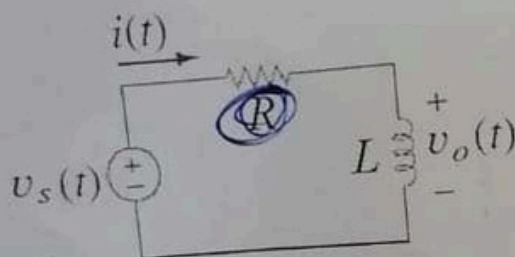


Fig. B.1

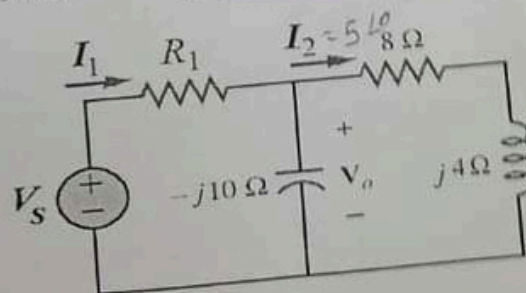


Fig. B.2

Consider the circuit shown in Fig. B.2, where $R_1 = 4 \Omega$.

4. The equivalent impedance (Z_{eq}) seen by the source is
 a. $12 - j4 \Omega$ b. $16 - j4 \Omega$ c. $10 - j4 \Omega$ d. $14 - j4 \Omega$
5. When $I_2 = 5/0^\circ \text{ A}$ then $I_1 =$
 a. $6 + j8 \text{ A}$ b. $3 + j4 \text{ A}$ c. $12 + j16 \text{ A}$ d. $9 + j12 \text{ A}$
6. When $V_o = 10/0^\circ \text{ V}$ then $V_s =$
 a. $18 + j4 \text{ V}$ b. $14 + j2 \text{ V}$ c. $16 + j3 \text{ V}$ d. $12 + j1 \text{ V}$

Consider the circuit shown in Fig. B.3, where $V_1 = 4/0^\circ \text{ V}$ and $V_2 = 3/0^\circ \text{ V}$. (Use nodal method)

7. The voltage across the capacitor (V_x) is
a. $5 + j1$ V b. $5 + j3$ V c. $5 - j3$ V d. $5 - j1$ V
8. The current (I_1) is
a. $0.3 + j0.3$ A b. $-0.3 - j0.3$ A c. $0.1 + j0.1$ A d. $-0.1 - j0.1$ A
- The current (I_2) is
a. $0.05 - j0.15$ A b. $0.1 - j0.05$ A c. $0.15 + j0.05$ A d. $0.2 + j0.15$ A

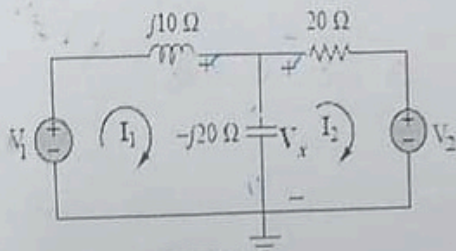


Fig. B.3

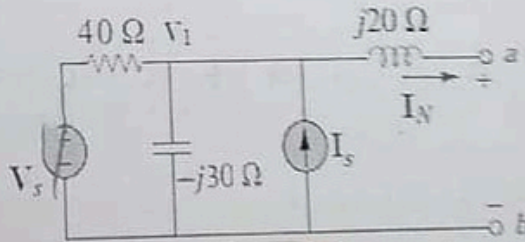
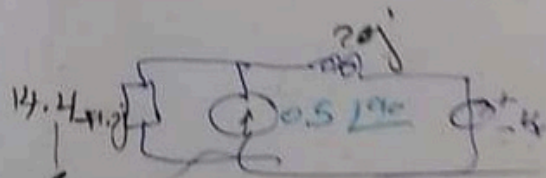
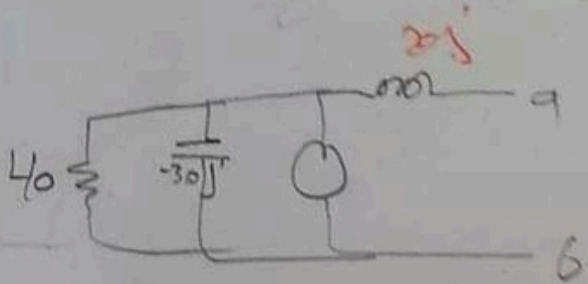


Fig. B.4

Consider the circuit shown in Fig. B.4, where $V_s = 15\angle 0^\circ$ V and $I_s = 0.5\angle 90^\circ$ A.

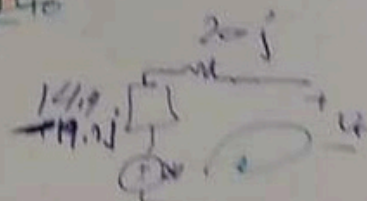
10. The voltage V_x due to the current source only
a. $14.4 + j10.8 \text{ V}$ b. $4.8 + j3.6 \text{ V}$ c. $19.2 + j14.4 \text{ V}$ d. $9.6 + j7.2 \text{ V}$
11. The voltage V_x due to the voltage source only
a. $9 - j12 \text{ V}$ b. $7.2 - j9.6 \text{ V}$ c. $5.4 - j7.2 \text{ V}$ d. $3.6 - j4.8 \text{ V}$
12. The value of V_x due to both sources
a. 15 V b. $42.4 - j12 \text{ V}$ c. $8.4 - j1.2 \text{ V}$ d. $21.6 - j1.2 \text{ V}$
13. The Thevenin equivalent impedance between the a - b terminals is
a. $14.4 + j5.8 \Omega$ b. $40 + j25 \Omega$ c. $j4 \Omega$ d. $14.4 - j19.2 \Omega$



24 $M = I \omega$

~~9.8 - 7.2~~

$$-21 + (14.4 + 0.8j) I + V_1 = 0$$



Consider the circuit shown in Fig. B.3, where $V_1 = 4/0^\circ$ V and $V_2 = 3/0^\circ$ V. (Use nodal method)

7. The voltage across the capacitor (V_x) is:
 a. $5 + j1$ V b. $5 + j3$ V c. $5 - j3$ V d. $5 - j1$ V
8. The current (I_1) is:
 a. $0.3 + j0.3$ A b. $-0.3 - j0.3$ A c. $0.1 + j0.1$ A d. $-0.1 - j0.1$ A
- The current (I_2) is:
 a. $0.05 - j0.15$ A b. $0.1 - j0.05$ A c. $0.15 + j0.05$ A d. $0.2 + j0.15$ A

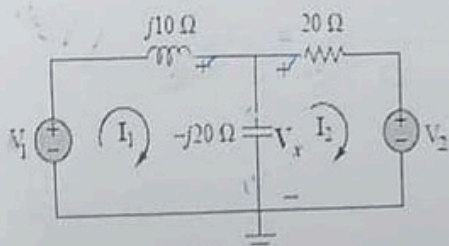


Fig. B.3

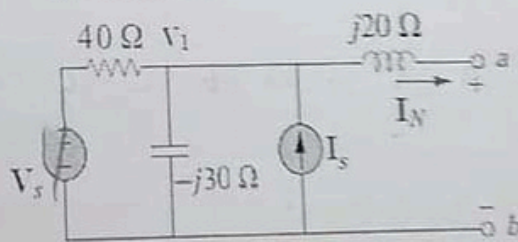


Fig. B.4

Consider the circuit shown in Fig. B.4, where $V_s = 15/0^\circ$ V and $I_s = 0.5/90^\circ$ A.

10. The voltage V_x due to the current source only:
 a. $14.4 + j10.8$ V b. $4.8 + j3.6$ V c. $19.2 + j14.4$ V d. $9.6 + j7.2$ V
11. The voltage V_x due to the voltage source only:
 a. $9 - j12$ V b. $7.2 - j9.6$ V c. $5.4 - j7.2$ V d. $3.6 - j4.8$ V
12. The value of V_x due to both sources:
 a. 15 V b. $42.4 - j12$ V c. $8.4 - j1.2$ V d. $21.6 - j1.2$ V
13. The Thevenin equivalent impedance between the $a-b$ terminals is:
 a. $14.4 + j5.8$ Ω b. $40 + j25$ Ω c. $j4$ Ω d. $14.4 - j19.2$ Ω

- Prob. 1: Consider the circuit shown in Fig. 1.
1. The load impedance (Z_L), in Ω , that absorbs the maximum average power is:
 2. The open circuit voltage between $a-b$ terminals (V_{oc}), when the load impedance is disconnected is:
 3. The short circuit current (I_{sc}), when $a-b$ terminals are shorted is:
 4. The maximum average power transferred to the load impedance (Z_L) is:



Fig. 1

(8 marks)

Prob. 2: Consider the circuit shown in Fig. 2 and using mesh analysis method,

1. Write the two mesh equations, do not simplify:

2. The current I_1 is:

3. The current I_2 is:

4. The capacitor current, I_C , is:

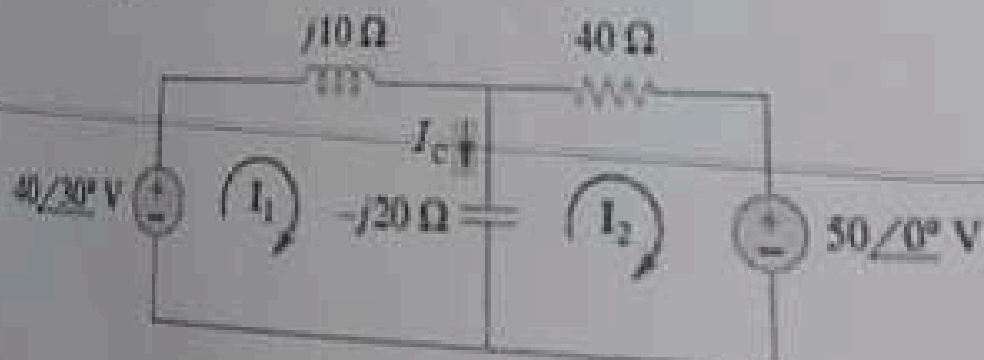


Fig 2

(8 marks)

Prob 3: Two loads are connected in parallel across a voltage source of $100\angle 0^\circ V_{\text{rms}}$, where load 1 absorbs 500VAR at 0.6 lagging p.f. while load 2 absorbs 400VA at 0.75 leading p.f.

1. The impedance of load 1 (Z_1), in Ω , is:
2. The current of load 2 (I_2), in Amps, is:
3. The complex power (S) supplied by the source is:
4. The power factor of the equivalent load is:
5. The capacitance needed to raise the pf to unity is:
6. The current supplied by the source, after the power factor is corrected, is:

(12 marks)

Prob. 4: Consider the circuit shown in Fig. 3,

1. The equivalent impedance seen by the source, Z_{eq} , is:
2. The current supplied by the source, I_1 , is:
3. The load current, I_2 , is:
4. The voltage across the primary winding, V_1 , is:
5. The average power absorbed by the 2Ω resistor is:

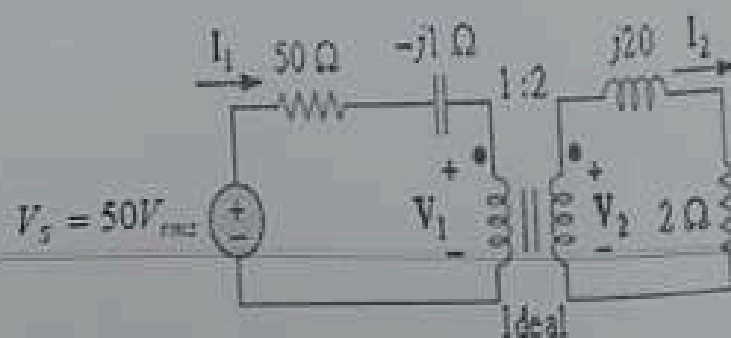


Fig.3

(10 marks)